# Constant-depth parity circuits versus superpolynomial monotone circuit size

Bruno P. Cavalar, Igor C. Oliveira

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*Allowed standard prerequisites (editorial).*

- Known monotone lower bounds, logspace algorithms and threshold circuit bounds. Exact author prior theorem statements retained with citations. Full author depth-reduction/padding support proofs included.

# 3 Preliminaries

## 3.1 Notation

**Boolean functions.** We denote by $\mathsf{Mono}$ the set of all monotone Boolean functions. We define $\mathsf{poly} = \left\{n \mapsto n^C : C \in \mathbb{N}\right\}$. A Boolean function $f : \{0,1\}^{\binom{n}{2}} \to \{0,1\}$ is said to be a graph property if $f(G) = f(H)$ for any two isomorphic graphs $G$ and $H$. Let $\mathcal{F} = \{f_n\}_{n\in\mathbb{N}}$ be a sequence of graph properties, where $f_n$ is defined over undirected graphs on $n$ vertices. We say that $\mathcal{F}$ is *preserved under homomorphisms* if, whenever there is a homomorphism from a graph $G$ to a graph $H$, we have $\mathcal{F}(G) \leq \mathcal{F}(H)$. We denote by $\mathsf{HomPreserving}$ the set of all graph properties which are preserved under homomorphisms. Note that $\mathsf{HomPreserving} \subseteq \mathsf{Mono}$.

**Boolean circuits.** We denote by $\mathsf{AC}^0_d[s]$ the family of Boolean functions computed by size-$s$, depth-$d$ Boolean circuits with unbounded fan-in $\{\wedge, \vee\}$-gates and input literals from $\{x_1, \overline{x_1}, \ldots, x_n, \overline{x_n}\}$. We write $\mathsf{AC}^0[s]$ as a shorthand for $\bigcup_{d=1}^{\infty} \mathsf{AC}^0_d[s]$, and $\mathsf{AC}^0$ as a shorthand of $\mathsf{AC}^0[n^{O(1)}] = \mathsf{AC}^0[\mathsf{poly}]$. We will also refer to $\mathsf{AC}^0_d[\mathsf{poly}]$ by $\mathsf{AC}^0_d$. We write $\mathsf{DNF}[s]$ to denote the family of Boolean functions computed by size-$s$ DNFs, where size is measured by number of terms. We write $\mathsf{CNF}[s]$ analogously. We write $\mathsf{SIZE}[s]$ to denote the family of Boolean functions computed by size-$s$ circuits. We write $\mathsf{DEPTH}[d]$ to denote the family of Boolean functions computed by fan-in 2 circuits of depth $d$. We denote by $\mathsf{AC}^0[\oplus]$ the family of Boolean functions computed by polynomial-size $\mathsf{AC}^0$ circuits with unbounded fan-in $\oplus$-gates.

We denote by $\mathsf{L}$ the family of Boolean functions computed by logspace machines, and by $\mathsf{NL}$ the family of Boolean functions computed by polynomial-time nondeterministic logspace machines. Moreover, we denote by $\oplus\mathsf{L}$ the family of Boolean functions computed by polynomial-time nondeterministic logspace machines with a *parity* acceptance condition (i.e., an input is accepted if the number of accepting paths is odd).

**Circuit complexity.** Given a circuit class $\mathcal{C}$, we write $\mathsf{m}C$ to denote the monotone version of $\mathcal{C}$. Given a function $f$, we write $\mathsf{mSIZE}(f)$ to denote the size of the smallest monotone circuit computing $f$ and $\mathsf{mDEPTH}(f)$ to denote the smallest depth of a fan-in 2 monotone circuit computing $f$. Given two Boolean functions $f, g$, we write $f \leq_m^{\mathsf{mProj}} g$ if there exists a many-one reduction from $f$ to $g$ in which each bit of the reduction is a monotone projection[1] of the input.

**Miscellanea.** Let $\alpha \in \{0,1\}^n$ . We define $|\alpha|_1 := \sum_{i=1}^n \alpha_i$. We call $|\alpha|_1$ the *Hamming weight* of $\alpha$. We let $\mathrm{supp}(\alpha) = \{i \in [n] : \alpha_i = 1\}$. We let $\mathrm{THR}_{k,n} : \{0,1\}^n \to \{0,1\}$ be the Boolean function such that $\mathrm{THR}_{k,n}(x) = 1 \iff |x|_1 \geq k$.

## 3.2 Background results

The next lemma, which is proved via a standard "guess-and-verify" approach, shows that nondeterministic logspace computations can be simulated by circuits of size $2^{n^\varepsilon}$ and of depth $d = O_\varepsilon(1)$.

**Lemma 3.1** (Folklore; see, e.g., [AHM+08, Lemma 8.1])**.** *For all $\varepsilon > 0$, we have $\mathsf{NL} \subseteq \mathsf{AC}^0[2^{n^\varepsilon}]$.*

**Theorem 3.2** ([GKRS19])**.** *There exists $\varepsilon > 0$ such that $\oplus\mathsf{L} \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$. Moreover, this separation is witnessed by* 3-XOR-SAT*.*

**Theorem 3.3** (Polynomial-size constant-depth vs. larger monotone size)**.** *For every $k \geq 1$, we have $\mathsf{AC}^0[\oplus] \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{(\log n)^k}]$.*

**Lemma 3.4** (Folklore; see, e.g., [AKR+01, OSS19])**.** *Let $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function computed by a $\oplus\mathsf{L}$ machine. For every $\delta > 0$, there exists an $\mathsf{AC}^0[\oplus]$ circuit of size $2^{n^\delta}$ that computes $f$.*

*Proof.* Let $M$ be a $\oplus\mathsf{L}$-machine computing $f$. Without loss of generality, we may assume that each configuration in the configuration graph $G$ of $M$ is *time-stamped* – in other words, each configuration carries the information of the number of computational steps it takes to arrive at it.[2] We may also assume that every accepting computation takes exactly the same amount of time, which means that every path from the starting configuration $v_{\mathsf{start}}$ to the accepting configuration $v_{\mathsf{accept}}$ has the same length in the configuration graph. These assumptions imply that the configuration graph is *layered* (because a configuration with time-stamp $t$ can only point to configurations with time-stamp $t+1$) and acyclic. Note that, for a fixed machine, the configuration graph can be computed from the input string using a projection.

Let $m = n^{O(1)}$ be the time that an accepting computation takes. We now show how to count (modulo 2) the number of accepting paths from $v_{\mathsf{start}}$ to $v_{\mathsf{accept}}$ with a depth-$d$ $\mathsf{AC}^0[\oplus]$ circuit. First, choose $m^{1/d}-1$ configurations $v_1, \ldots, v_{m^{1/d}-1}$ (henceforth called "checkpoints") from $V(G)$, such that the configuration $v_i$ is at the level $i \cdot m^{1-1/d}$ in the configuration graph (i.e., it takes $i \cdot m^{1-1/d}$ time steps to arrive at $v_i$). For convenience, we let $v_0 = v_{\mathsf{start}}$ and $v_{m^{1/d}} = v_{\mathsf{accept}}$. We then count the number of paths from from $v_{\mathsf{start}}$ to $v_{\mathsf{accept}}$ that go through $v_1, \ldots, v_{m^{1/d}-1}$, and sum over all possible choices of the checkpoints. Since the graph is layered and each path from $v_0$ to $v_{m^{1/d}}$ has length exactly $m$, there is only one choice of checkpoints that witnesses a given path from $v_0$ to $v_{m^{1/d}}$, so no path is counted twice in this summation. Letting $\#\mathsf{paths}(s,t,\ell)$ denote the number of paths between configurations $s$ and $t$ with distance exactly $\ell$, we obtain

$$\#\mathsf{paths}(v_0, v_{m^{1/d}}, m) = \sum_{v_1,\ldots,v_{m^{1/d}-1}} \prod_{i=0}^{m^{1/d}-1} \#\mathsf{paths}(v_i, v_{i+1}, m^{1-1/d}).$$

The above calculation can be done in modulo 2 with an unbounded fan-in XOR gate (replacing the summation) and an unbounded fan-in AND gate (replacing the product). Note that the formula above is recursive. Repeating the same computation for calculating (modulo 2) the expression $\#\mathsf{paths}(v_i, v_{i+1}, m^{1-1/d})$ for each $i$, we obtain a depth-$2d$ $\mathsf{AC}^0[\oplus]$ circuit for calculating the number of paths from $v_{\mathsf{start}}$ to $v_{\mathsf{accept}}$ (modulo 2). Clearly, the total size of the circuit is $2^{O(m^{1/d} \cdot \log m)}$, which is smaller than $2^{n^\delta}$ for a large enough constant $d$. □

[1] A monotone projection is a projection without negations.

[2] Formally, we can define a $\oplus\mathsf{L}$-machine $M'$ such that the configurations of $M'$ are $(C,t)$, where $C$ is a configuration of $M$, and $t = 0, 1, \ldots, m = n^{O(1)}$ is a number denoting the time in which the configuration was achieved. A configuration $(C,t)$ can only reach a configuration $(C', t+1)$ in the configuration graph of $M'$.

*Proof.* By Theorem 3.2, there exists $\varepsilon > 0$ and a monotone function $f \in \oplus\mathsf{L}$ such that any monotone circuit computing $f$ has size $2^{\Omega(n^\varepsilon)}$.

Let $\delta = \varepsilon/k$ and let $m = 2^{n^\delta}$. Let $g : \{0,1\}^n \times \{0,1\}^m \to \{0,1\}$ be the Boolean function defined as $g(x,y) = f(x)$. Note that $g$ is a function on $N := m + n = 2^{\Theta(n^\delta)}$ bits. By Lemma 3.4, there exists an $\mathsf{AC}^0[\oplus]$ circuit computing $f$ of size $2^{n^\delta} = N^{O(1)}$. The same circuit computes $g$. On the other hand, any monotone circuit computing $g$ has size $2^{\Omega(n^\varepsilon)} = 2^{\Omega((\log N)^{\varepsilon/\delta})} = 2^{\Omega((\log N)^k)}$. □

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